

Lecture “Stereochemistry”

Take-home messages 1

1. Nomenclature

Isomers: Constitutional isomers, stereoisomers, geometric isomers, diastereomers, enantiomers, epimers, meso-compounds

Chirality: Stereocenters, axial chirality, planar chirality, helical chirality, topological chirality, CIP-Rules: R/S, like/unlike, Re/Si, P/M

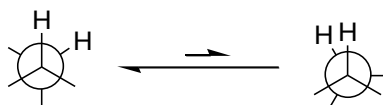
Topicity: homotopic, enantiotopic, diastereotopic

Reactions: Enantioface-differentiating, enantiotopos-differentiating, enantiomer-differentiating

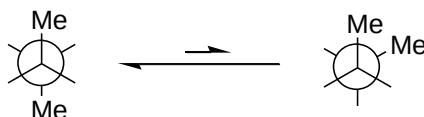
2. Conformational Analysis

Alkanes: - 3 important factors:

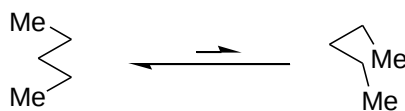
- staggered better than eclipsed (ethane)



- antiperiplanar better than gauche by 0.9 kcal (butane)

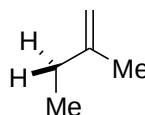


- double gauche pentane very bad (pentane)



Alkenes: - 2 important factors:

- A^{1,2} – strain



- A^{1,3} – strain

